

MAGNETOM

Installation and Startup System

UPS Vertiv Liebert GXT-4

Vida, Sola, Terra, Prisma,
Prisma fit, Aera, Skyra,
Avanto fit, Skyra fit,
Avanto, Verio, Espree,
SATS, TATS, Sonata, Trio
ESSENZA, Spectra, Amira,
Sempra, MAGNETOM C!

Document Version

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1.1 Safety Information

For your safety and the safety of third parties as well as for protecting the MR system and its components from damage, the appropriate safety information is included in register "General Safety Notes" and "MR-Specific Safety Information" in the "Technical Documents".

Prior to starting work, read the specific information in the respective manual:

- General safety information for service personnel.
- Limitations with respect to ferromagnetic materials.
- Safety information pertaining to working with the magnet at field.
- Information regarding working with heavy loads.
- Protective conductor test before switching on the system and for subsequent repairs.

1.2 General

The documentation describes the installation of the UPS Vertiv Liebert GXT-4.

1.2.1 Function

Using a UPS (uninterruptable power supply) ensures that the system computer is protected against any interruption of the power supply.

In most cases, local voltage supplies are sufficient for satisfactory operation.

However, modern processor-controlled configurations require highly reliable sources. The Vertiv Liebert GXT-4 meets these requirements. The Vertiv Liebert GXT-4 is the standard UPS for the **MRAWP Host PCs** and the **MaRS** (at MAGNETOM Terra in a separate cabinet).



The Vertiv Liebert GXT-4 including the Siemens cable set supports the MRAWP Host PC and the MaRS.

For other options, such as the MRWP, procure an additional UPS system, preferably a Vertiv Liebert GXT-4.

1.2.2 Technical dData

Technical data of Vertiv Liebert GXT-4	
Weight in kg	32.4
Dimensions of housing H x W x D in mm	602 x 85 ¹ x 430 ²
Output in kVA	3.0
Backup time in min.	at max. load 5 min. with built-in batteries
Temperature	+0 °C to +40 °C
Relative humidity	5 % - 95 % noncondensing
Acoustic noise	< 48 dBA ³

1. 215 mm with feet
2. 440 mm with feet
3. at a distance of 1.0 meter

1.2.3 UPS System for the Host PC



If an older UPS with a distribution box was used, it must always be replaced with a Vertiv Liebert GXT-4 UPS.

The old UPS and the distribution box need to be removed.



When installing a MRWP console, the UPS Vertiv Liebert GXT-4 does not support the MRWP PC. You must install a standard PC UPS or an additional UPS Vertiv Liebert GXT-4 (without a cable set) to supply the MRWP PC. The UPS Vertiv Liebert GXT-4 is used for the MRAWP Host computer.

1.2.4 Required tools and auxiliary tools

- Tool kit

1.2.5 Time required

Installation	approx. 2 hr.
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1.2.6 Items included in the shipment



Check the parts using the material list supplied with the delivery volume.

Tab. 1 Vida - scope of delivery with UPS Vertiv Liebert GXT-4:

Material no.	Contents
112 91 558 (Vida)	UPS Vertiv Liebert GXT-4
	Power cable 9m (W10610)
	Status cable 9m (W10611)
	Plastic tower stand sets
	User manual

Tab. 2 Scope of delivery with UPS Vertiv Liebert GXT-4 :

Material no.	Contents
112 91 557 (all other systems)	UPS Vertiv Liebert GXT-4
	Power cable 9m (W10610)
	Plastic tower stand sets
	User manual



For MAGNETOM Terra a second UPS Vertiv Liebert GXT-4 has to be ordered!
The second UPS is used for the MaRS cabinet.

Fig. 1: UPS Vertiv Liebert GXT-4



Fig. 2: Power cable W10610



Fig. 3: Status cable W10611 (only for Vida)



1.2.7 Battery Module (optional)

Tab. 3 Battery module (optional):

Material no.	Contents
112 91 559	Battery module
	Cable (0.65 m)

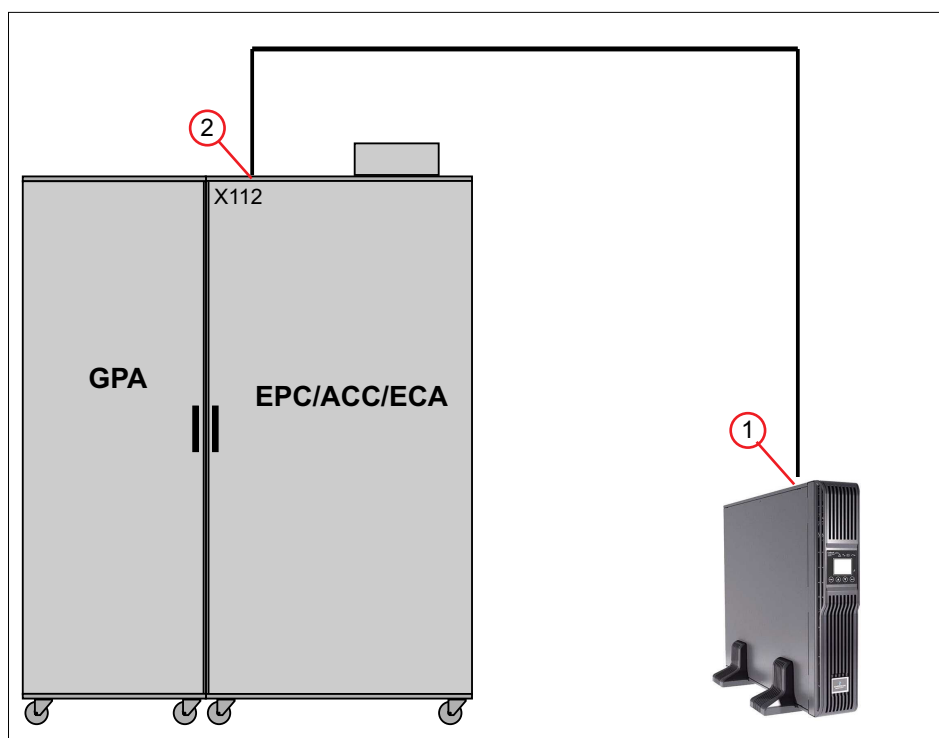


Optional battery cabinets are available for the UPS Vertiv Liebert GXT-4 . For the installation steps, refer to (→ Installing the Battery Cabinet (optional) / Page 18).

2.1 Overview

2.1.1 Overview

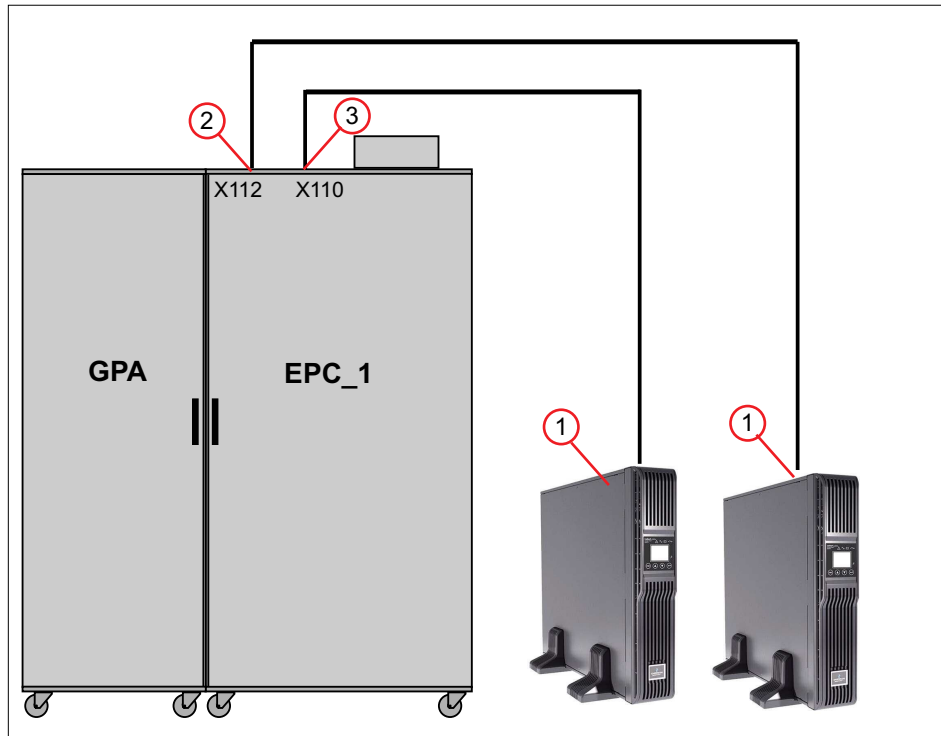
Fig. 4: UPS configuration for the host computer



- (1) UPS Vertiv Liebert GXT-4
- (2) To EPC/ACC/ECA X112; CCA X103

2.1.2 Overview MAGNETOM Terra

Fig. 5: UPS configuration for MAGNETOM Terra



2.2 Installation



The Vertiv Liebert GXT-4 output voltage is configured to 230 V for delivery.

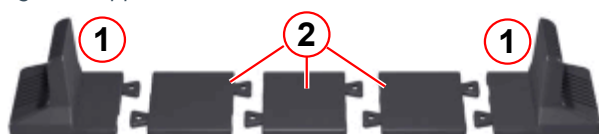
2.2.1 Mechanical installation

Fig. 6: Front plastic bezel cover



- Place the front plastic bezel cover on the UPS Vertiv Liebert GXT4.

Fig. 7: Support bases for tower installation



- (1) Support bases
(2) Spacers

- Connect the spacers and the support bases as shown in Fig. 4.
- If optional Liebert external battery cabinets will be connected to the Vertiv Liebert GXT4, take out the spacers delivered with the battery cabinet.

Fig. 8: Tower installation



- (1) Support bases

- The UPS Vertiv Liebert GXT4 needs two assembled support bases, one in the front and one in the rear.

2.2.2 Preparation



For additional information, please refer to the documentation for the components of the UPS Vertiv Liebert GXT-4 supplied with the system.

- Shutting down the software, no new start.
- Switch off the system.
- Switch off the circuit breakers:

Harmony/Symphony family:

- F23 in the mains box on top of the CCA.

Tim Systems:

- F100 in the mains box on top of the ACC.

Vida/Terra/Aera/Skyra/Avanto^{fit}/Skyra^{fit}/Prisma/Prisma^{fit}/Amira/Sempra/Spectra:

- F100 in the mains box on top of the EPC.

Essenza, C2!:

- F100 in the mains box on top of the ECA.
- Check to ensure that voltage is no longer present.



Place the UPS Vertiv Liebert GXT-4 next to the EPC/ACC/CCA/ECA (equipment room) and make sure that the airflow around the unit is unrestricted.

2.2.3 Cabling

Fig. 9: UPS Vertiv Liebert GXT-4, rear view

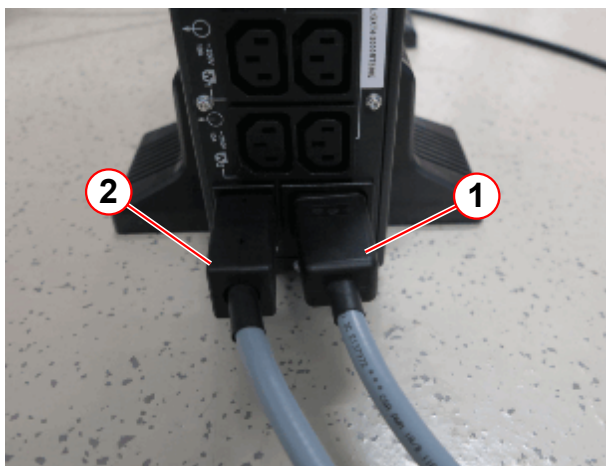


- (1) Output power
- (2) Input power
- (3) Communication terminals 1-8

Connecting status cable W10611 to the communication terminals at the UPS Vertiv Liebert GXT-4

1 + 2	REPO
5 + 6	Power loss (only for Vida)
7 + 8	Battery low (only for Vida)

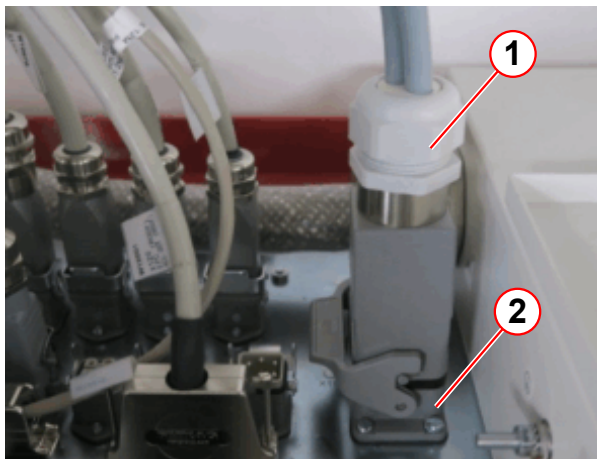
Fig. 10: Power cable (W10610)



- (1) Power output
- (2) Power input

- Connect the two plugs of the power cable W10610 to the power in and output on the UPS Vertiv Liebert GXT-4. See Fig. 7.
(→ Fig. 10 Page 12)

Fig. 11: EPC roof - X112



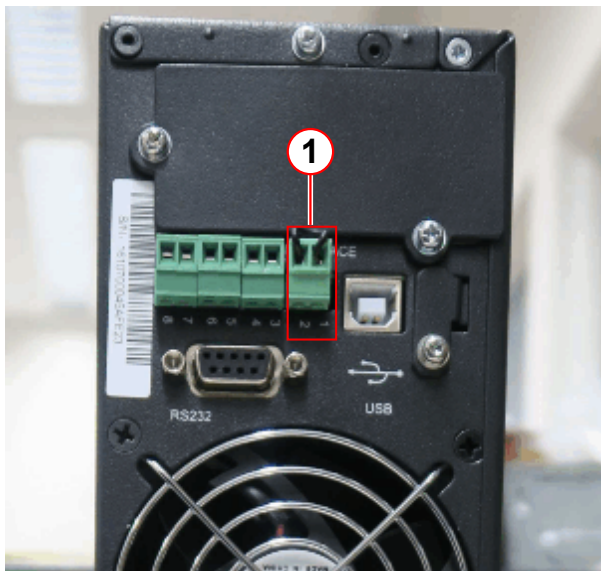
- (1) Power cable W10610
(2) X112 on EPC roof

- Connect the other end of power cable W10610 to:
 - X112 on top of the EPC/ACC/ECA
 - X103 on top of the CCA/ECA

Dependent to the available system:

- **Only for MAGNETOM Terra:**
Connect the power cable W10610 of the second UPS Vertiv Liebert GXT-4 to:
 - X110 on top of the ECP

Fig. 12: Communication terminals 1+2



- (1) REPO contact

- Connect the local EPO circuit to communication terminals 1+2 (REPO contact) of the UPS Vertiv Liebert GXT-4.

Dependent to the available system:

- **Only for MAGNETOM Terra:**
The REPO contacts of **both** UPS systems have to be connected in **parallel** (pay attention to the polarity!) to the EPO circuit of the main MR system!



If two or more UPS units are installed, the REPO contacts of the UPS need to be connected in parallel.

Pay attention to the correct polarity, connect the same REPO socket of each UPS to the same connection point of the EPO circuit.

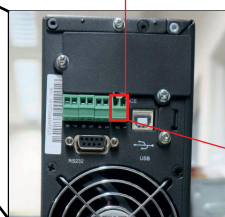
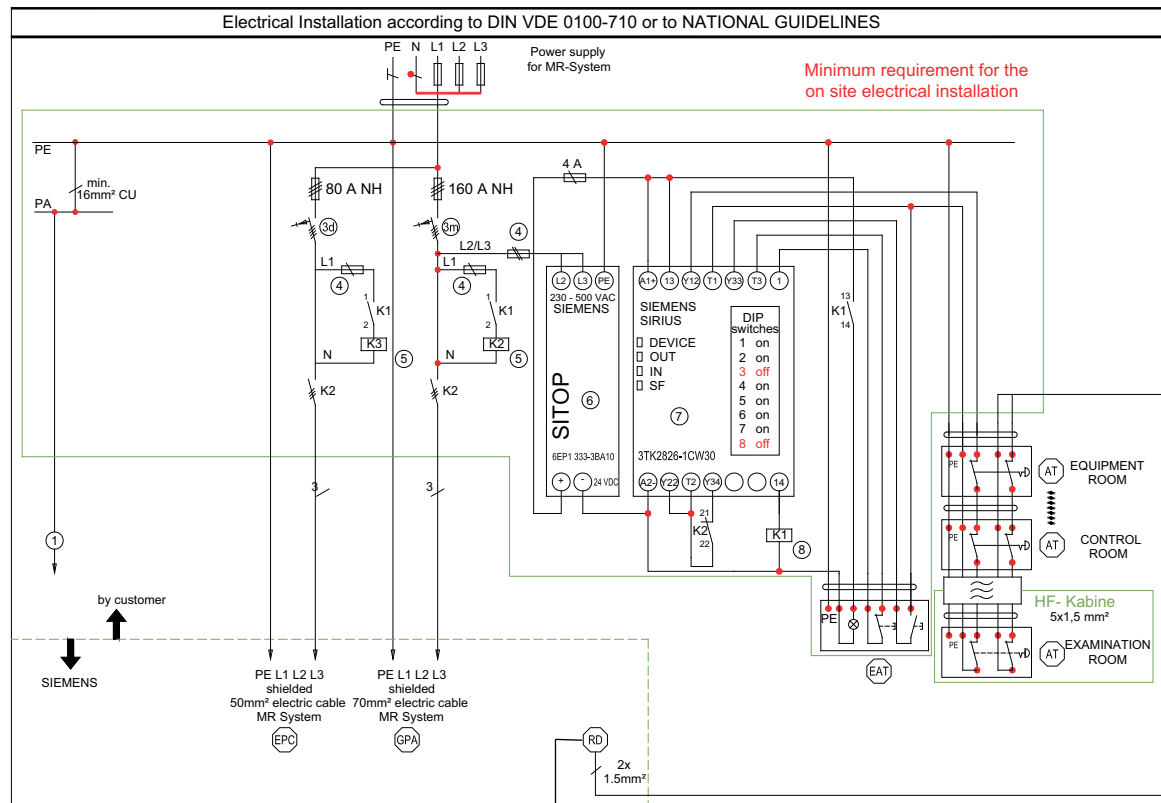
2.2.4 Connecting the REPO of the UPS



If a small UPS (for the host PC) is supplying the system, pressing the EPO button interrupts the power supply between the UPS and the EPC cabinet.

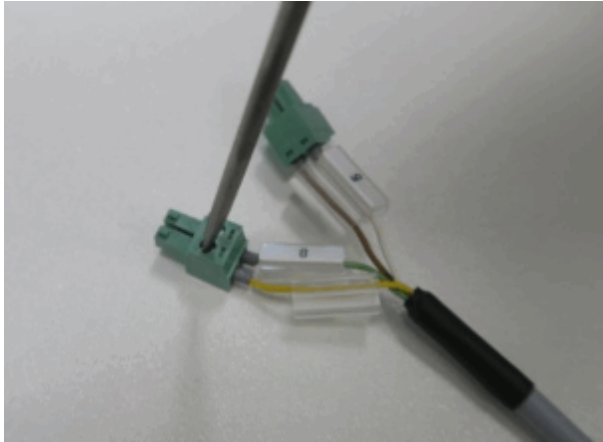
2.2.4.1 All systems

Fig. 13: Connecting the EPO



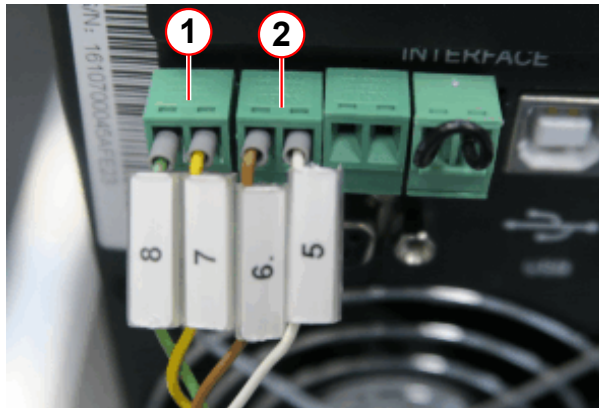
2.2.5 Only for MAGNETOM Vida

Fig. 14: Status cable W10611



- Connect the status cable W10611 to the plug connectors.
- Check the labels on the cable ends and make sure to connect the cables accordingly.

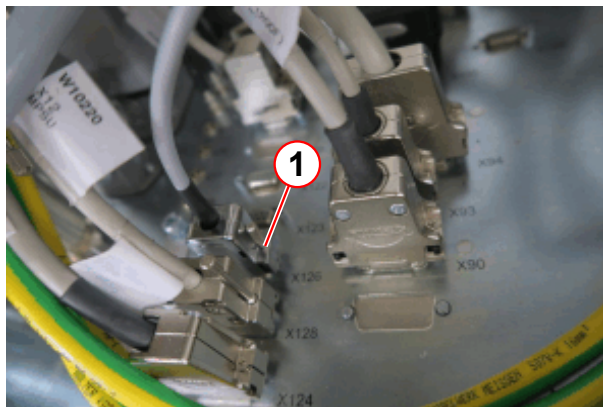
Fig. 15: Communication terminals 5+6 / 7+8



- (1) Battery low
(2) Power loss

- Connect the status cable W10611 to the communication terminals 5 + 6 and 7 + 8 of the UPS Vertiv Liebert GXT-4.
- Check the labels on the cable ends and make sure to connect the cables accordingly.

Fig. 16: EPC roof - X126



- (1) X126 on EPC roof

- Connect the other end of the status cable W10611 to X126 on the EPC roof.

2.3 Startup



The Vertiv Liebert GXT-4 output voltage is configured to 230 V for delivery.

Fig. 17: UPS power on



- Switch on the circuit breakers: F100 in the line power box on the top of the EPC.
- After the system is switched on, the UPS will boot up automatically.
- After the system has successfully booted, power on the UPS by selecting "Yes," and then pressing Enter.



It is important to power up and activate the UPS system in order to guarantee trouble-free system operation.

2.3.1 Final steps - Emergency Power Off Switches



The UPS system and the MR system must be switched off when the EPO switch (Emergency Power Off) is activated.

2.3.1.1 Checking the emergency power-off switches (EPO)

Prerequisites:

The system contactor in the EPC and GPA cabinet is switched on.

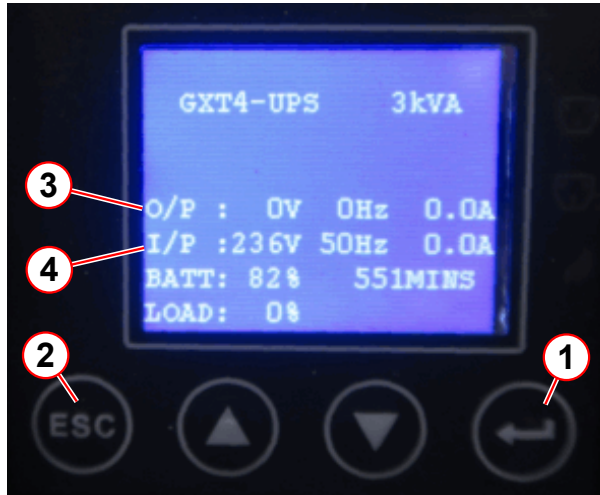
If the UPS is installed, the UPS display shows O/P : 230 V. Press the UPS Enter button (→ Fig. 18 Page 17) to open the display.

The system is not booted (system shutdown or press **F12** during boot).

- Press the emergency power off switch (EPO).
- Check to make sure that the electronics cabinet is de-energized.
 - » LEDs on the EPC cabinets are off.
- Check to make sure that the **GPA** cabinet is de-energized.
 - » LEDs on the GPA cabinets are off.

- Check the UPS output voltage, it must be switched off.
 - » The UPS display shows UPS Shutdown Command Received for a few seconds, and then O/P: 0 V (output voltage is 0 V). (→ Fig. 18 Page 17)

Fig. 18: UPS display



- (1) Enter button
- (2) ESC button
- (3) O/P output power
- (4) I/P input power

Release the EPO switch and reactivate the system by switching on the main power.

After the EPO switch is released, the UPS output power must be switched on manually.

- Switch on the UPS output power: navigate in the UPS display menu to switch on output power > press **Enter** button > **3_CONTROL** > **1_TURN ON & OFF** > **TURN ON UPS** > **Do you want to turn on UPS** > **YES** > **Enter**
 - » UPS output voltage is on (O/P: 230 V)
- **Perform this test on all emergency power-off switches in the equipment room, control room, and examination room.**



For MAGNETOM Terra perform the emergency power-off switches (EPO) test and check that both UPS Vertiv Liebert GXT-4 are switched off.

2.4 Installing the Battery Cabinet (optional)



For additional information, please refer to the UPS Vertiv Liebert GXT-4 documentation (supplied with the UPS).

Fig. 19: Battery cabinet

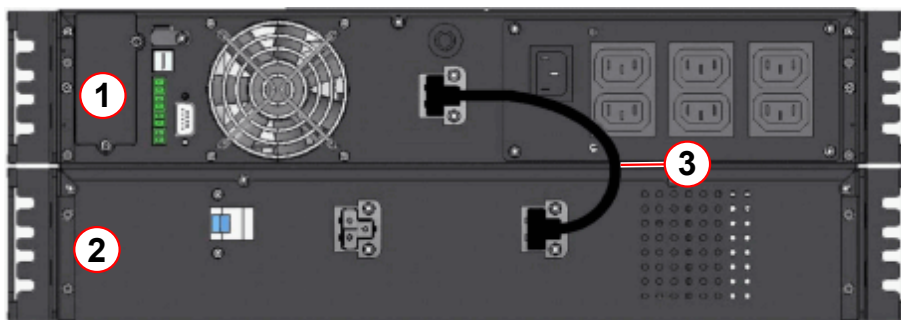


- (1) Isolation breaker
- (2) External battery connector

Connecting the battery cabinet:

- Verify that the battery isolation breaker is in the Off (open) position.
- Take out the battery cable included with the battery cabinet.
- Connect one end of the battery cable to the external battery connector on the rear panel of the UPS, and connect the other end to any battery port on the rear panel of the battery cabinet.
- Repeat steps 1 through 3 for each battery cabinet that will be connected to the system. Additional battery cabinets will plug into the previously installed battery cabinet.
- Switch on the battery breaker on the rear of the external battery cabinet.
- Use the configuration program included with the UPS to specify the number of external battery cabinets connected to the UPS Vertiv Liebert GXT-4.

Fig. 20: Battery cable attachment



- (1) UPS Vertiv Liebert GXT-4
- (2) External battery cabinet
- (3) Battery cable

Version 02

- MAGNETOM Terra was added.

Version 03

- Information for Biograph mMR were deleted.
- Sola was added.

There are no Hazard IDs in this document.

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